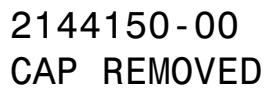
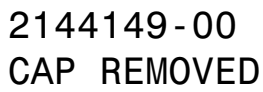
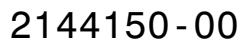
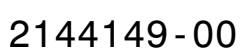
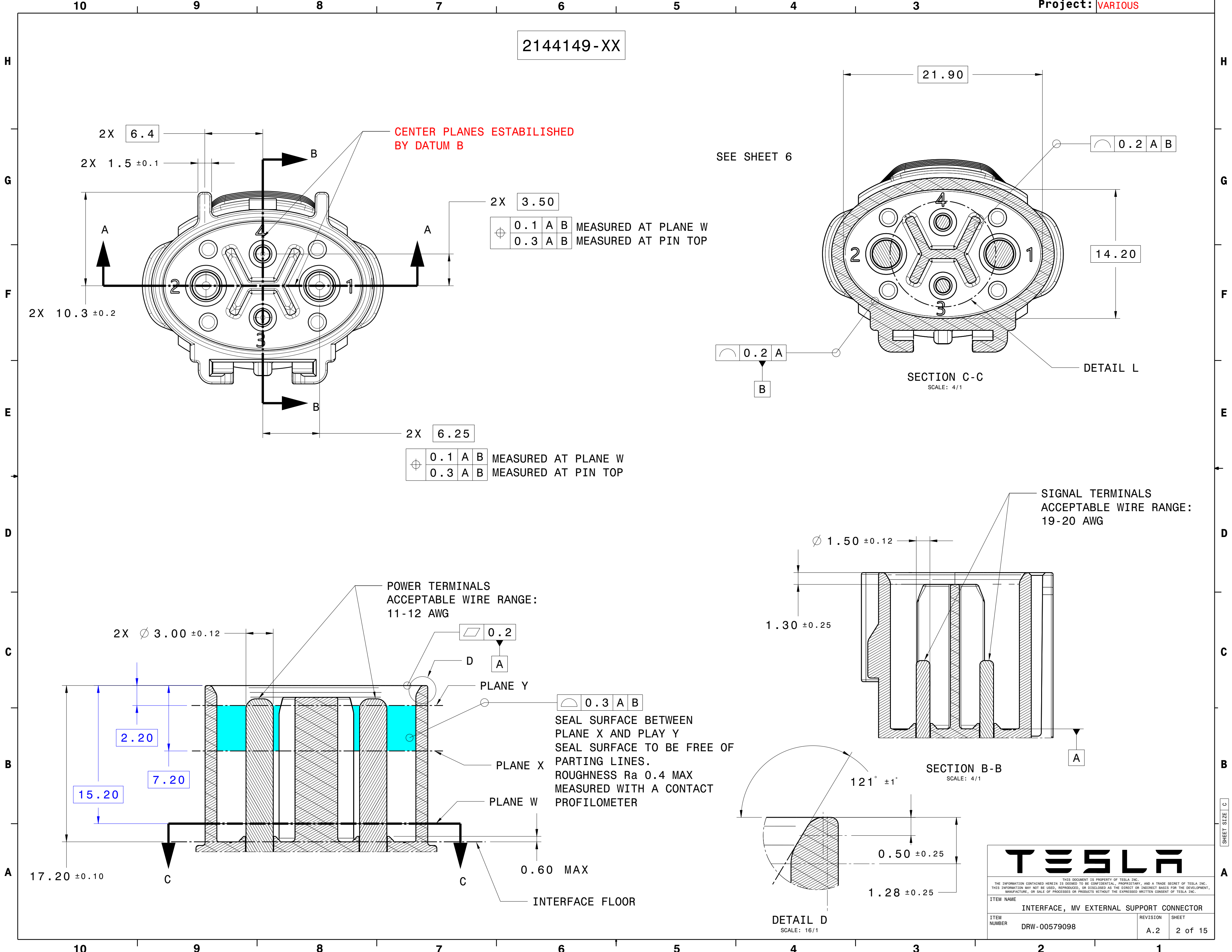


FOR COMPONENTS, ACCESSORIES, AND PACKAGING SIZE, REFER TO APPENDIX SHEETS

01



2144149-XX

Project: VARIOUS

CENTER PLANES ESTABLISHED
BY DATUM B

SEE SHEET 6

SECTION C-C
SCALE: 4/1

DETAIL L

POWER TERMINALS
ACCEPTABLE WIRE RANGE:
11-12 AWG

SIGNAL TERMINALS
ACCEPTABLE WIRE RANGE:
19-20 AWG

SEAL SURFACE BETWEEN
PLANE X AND PLAY Y
SEAL SURFACE TO BE FREE OF
PARTING LINES.
ROUGHNESS Ra 0.4 MAX
MEASURED WITH A CONTACT
PROFILOMETER

SECTION B-B
SCALE: 4/1

DETAIL D
SCALE: 16/1

INTERFACE FLOOR

TESLA

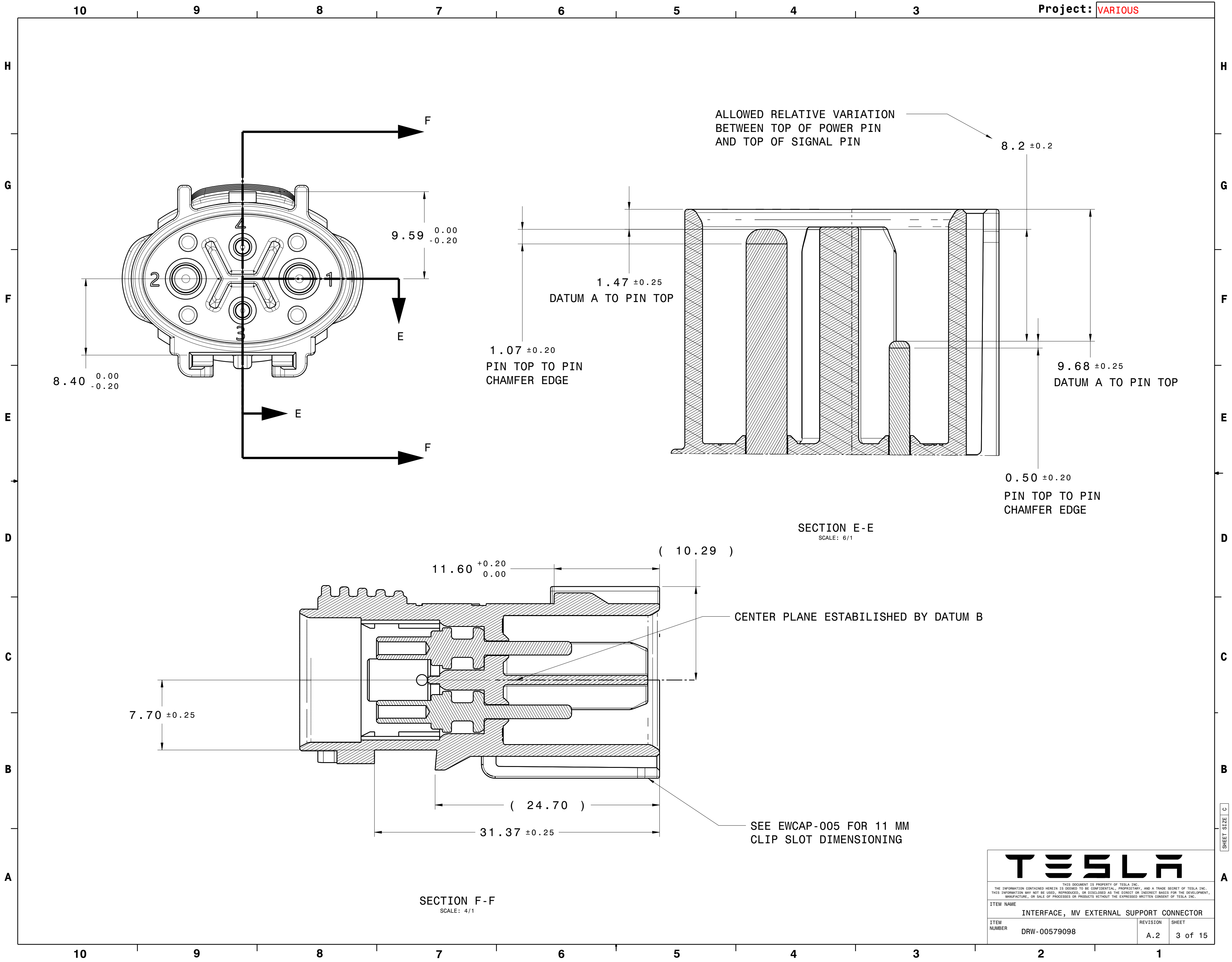
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ITEM NAME
INTERFACE, MV EXTERNAL SUPPORT CONNECTOR

ITEM NUMBER
DRW-00579098

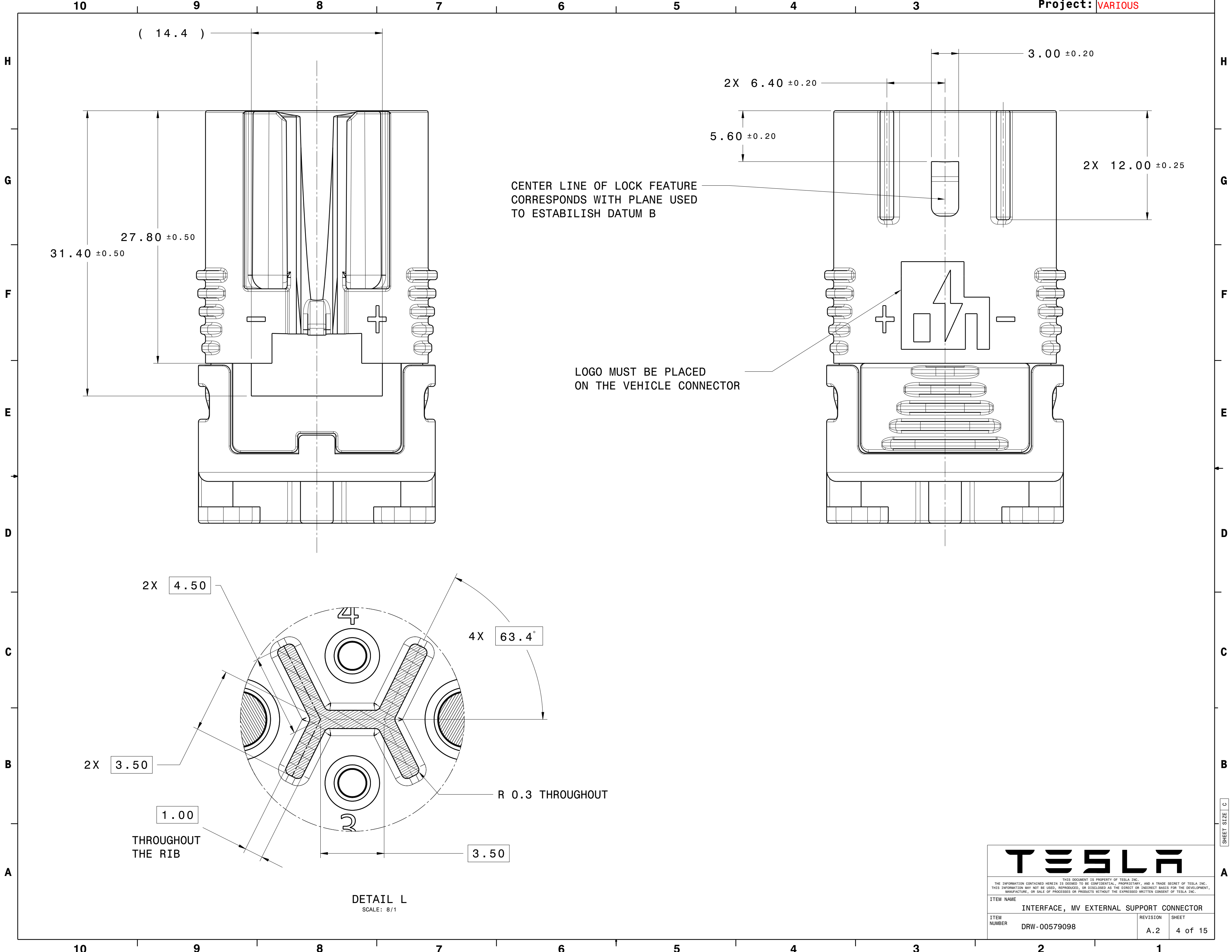
REVISION
A.2

SHEET
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ITEM NAME		
INTERFACE, MV EXTERNAL SUPPORT CONNECTOR		
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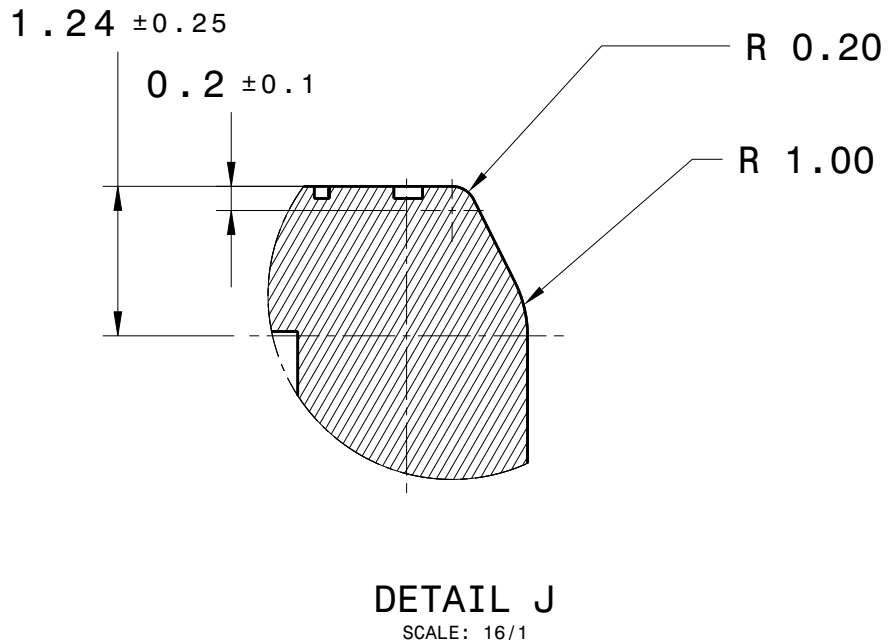
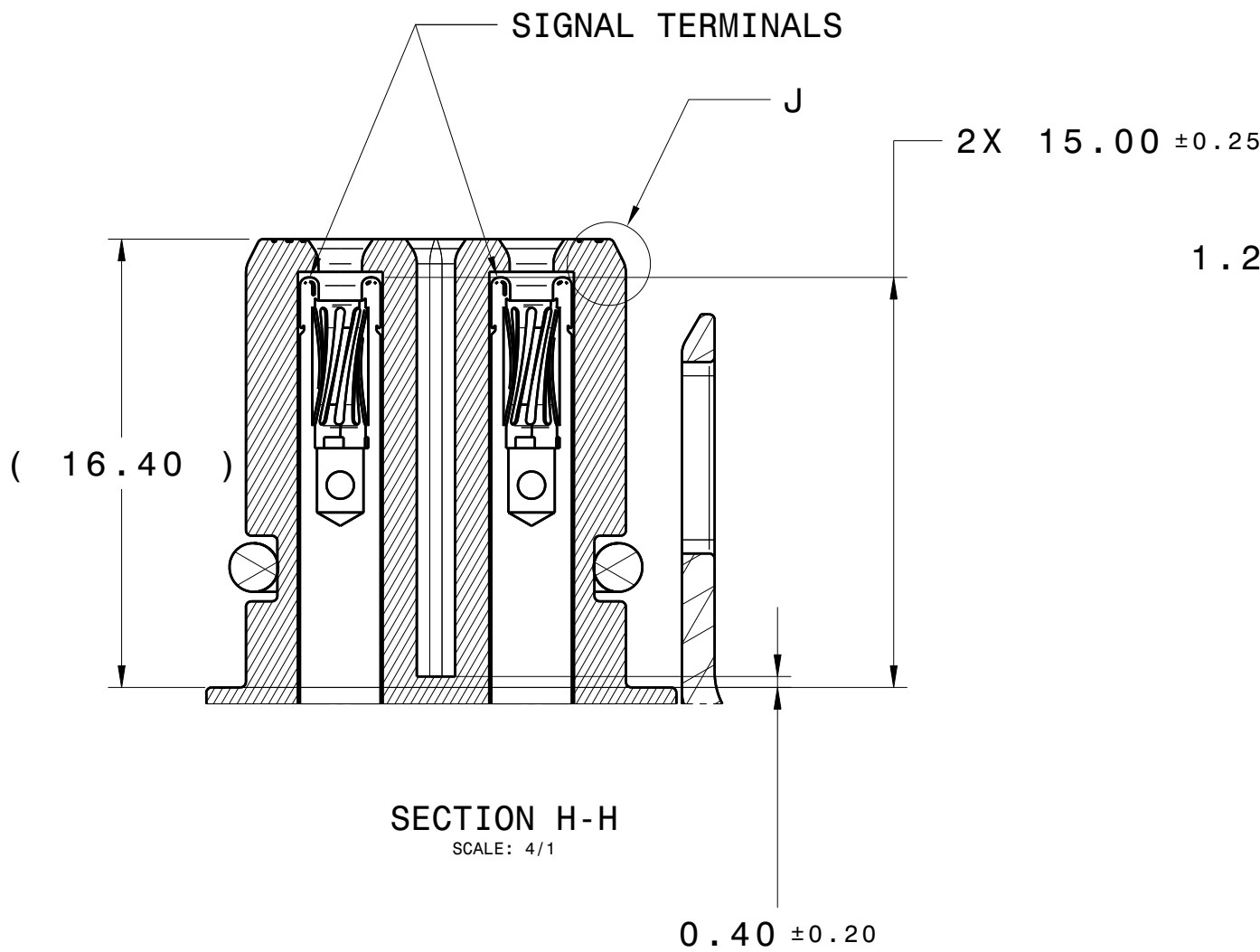
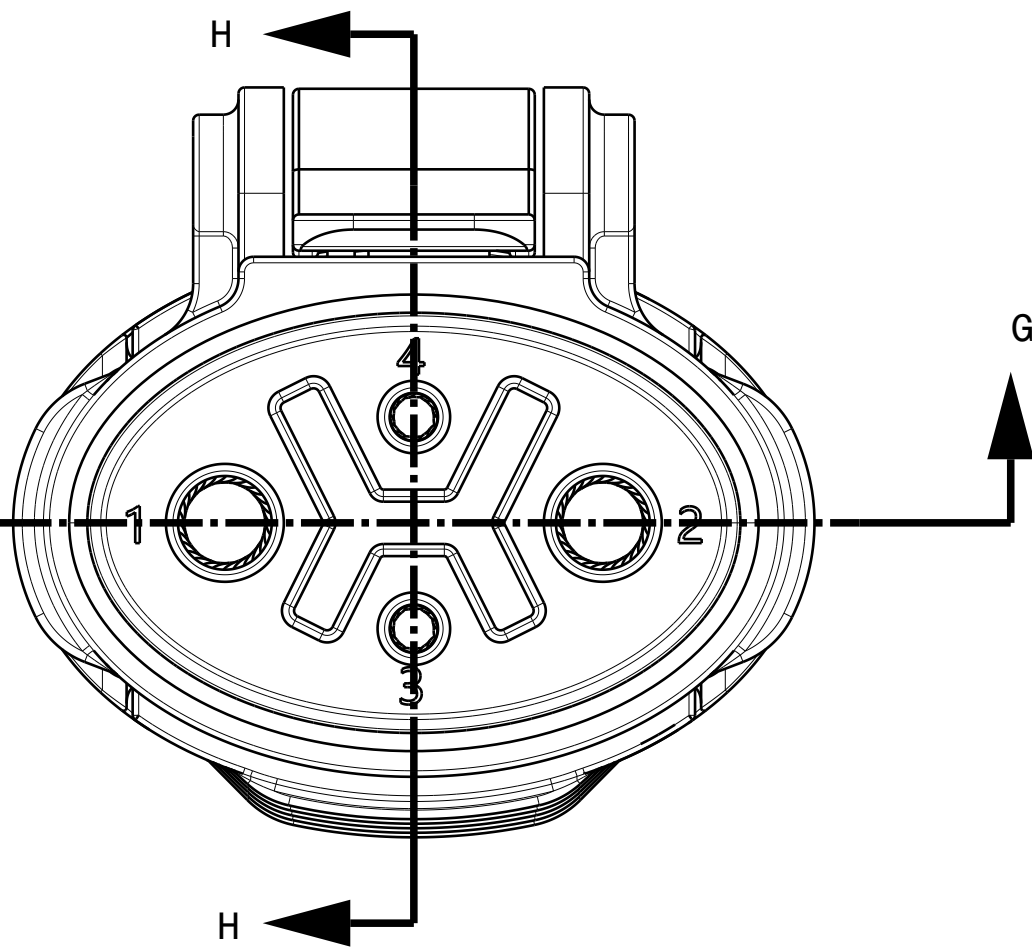


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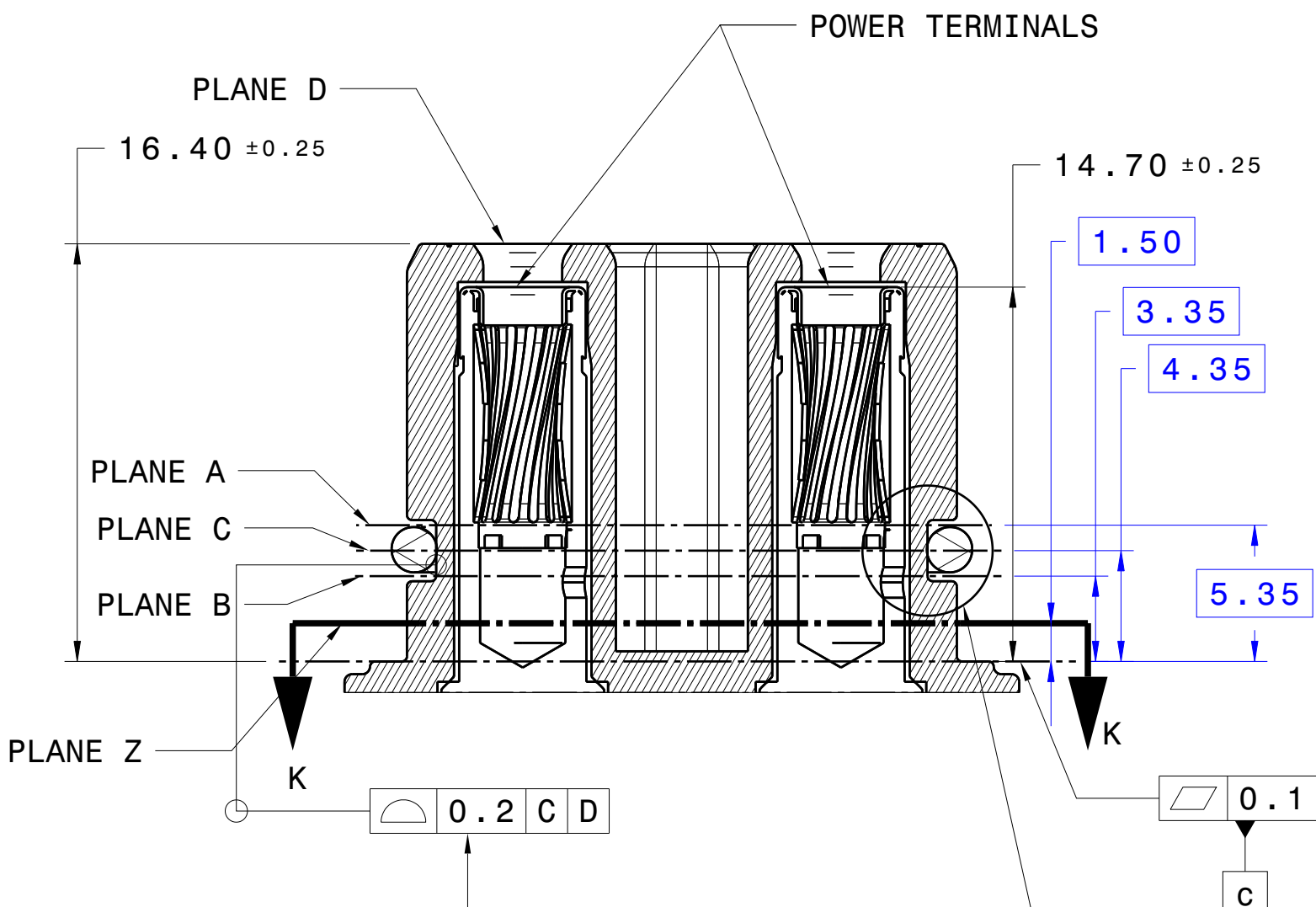
ITEM NUMBER	REVISION	SHEET
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2144150-XX



SECTION H-H
SCALE: 4/1

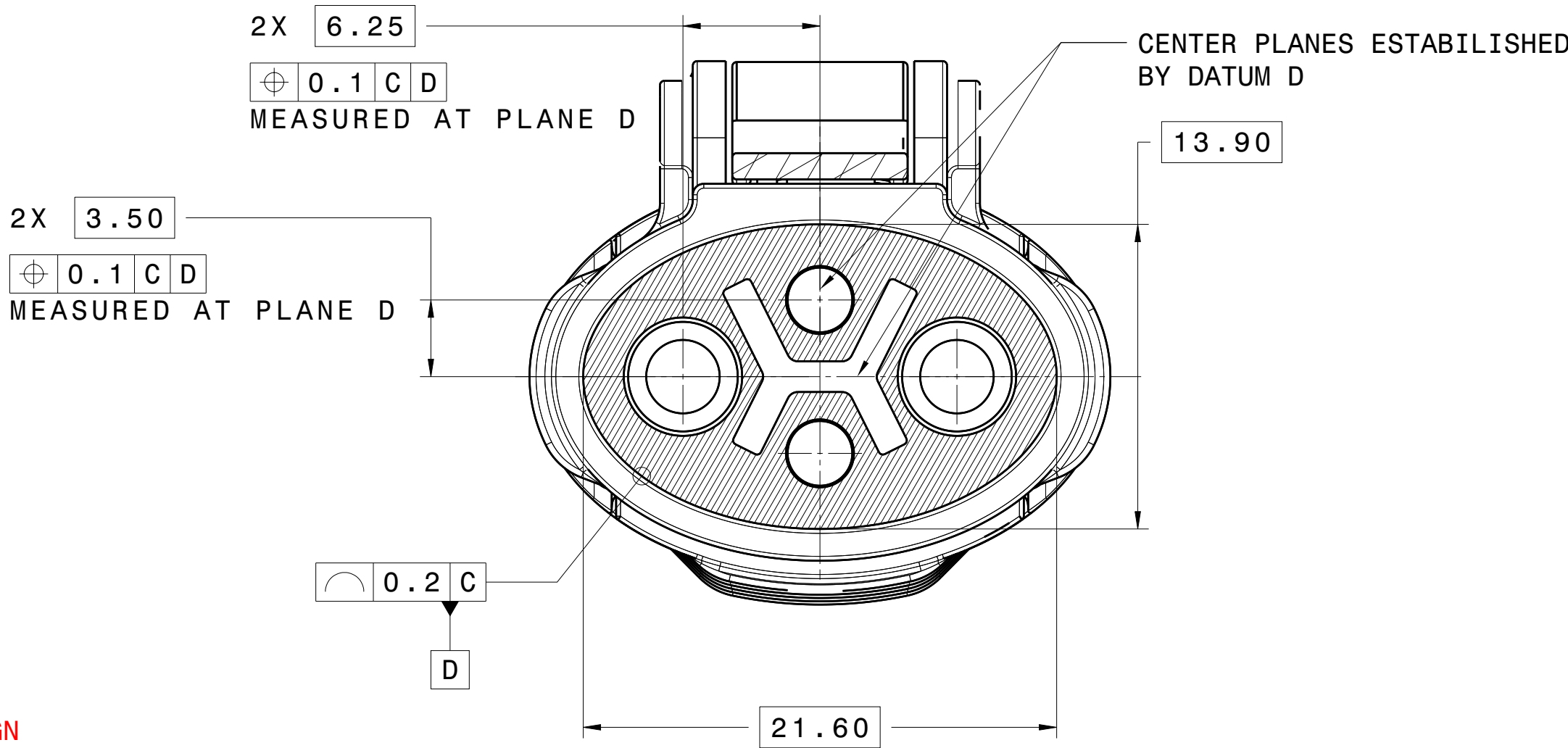
DETAIL J
SCALE: 16/1



O-RING GLAND DESIGN
PER SAE AS5857 SIZE 015
CENTER PLANE OF O RING SHOULD COINCIDE
WITH PLACE

APPLIES TO SEALING SURFACE
BETWEEN PLANE A AND PLANE B
SEALING SURFACE TO BE
FREE OF PARTING LINES

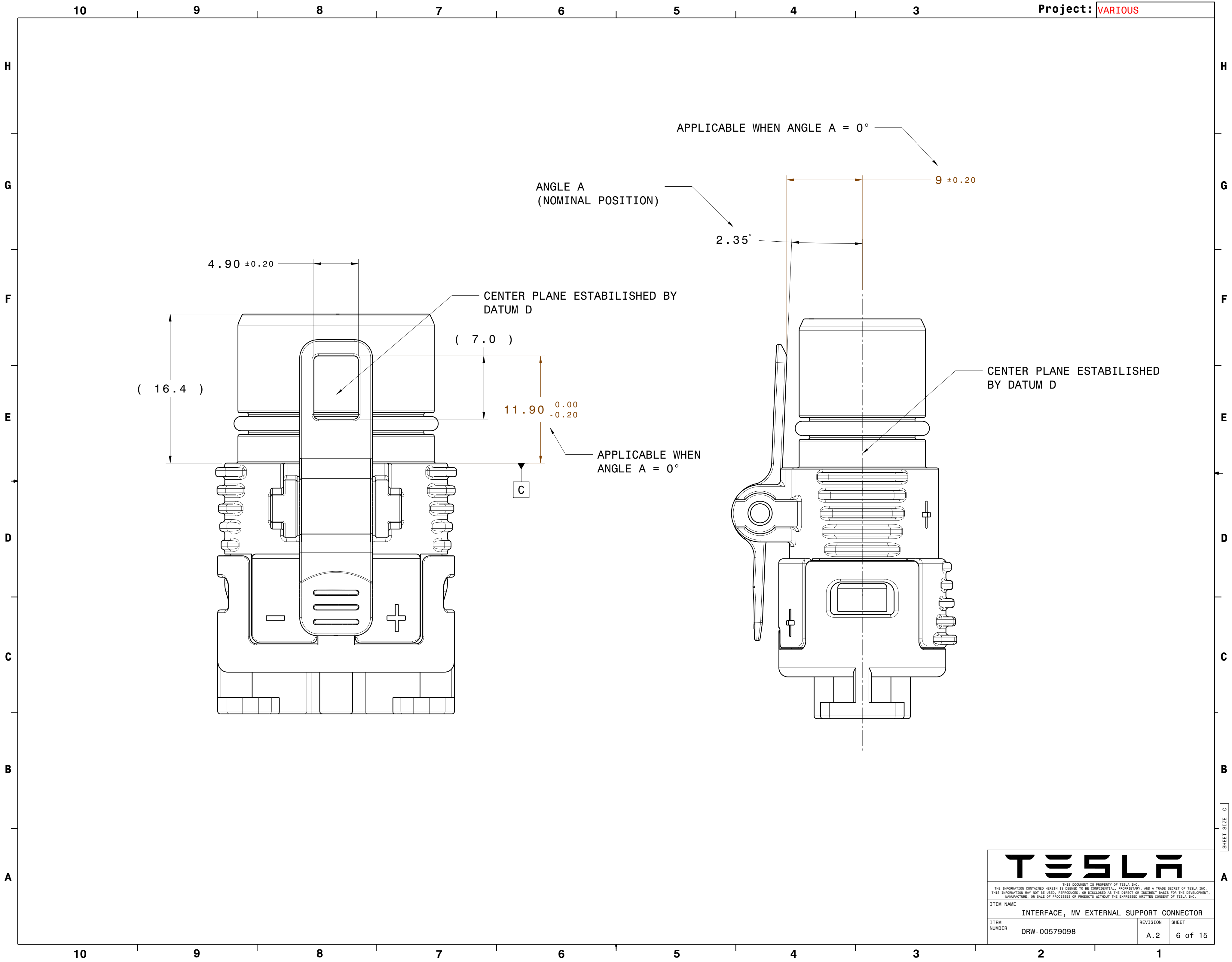
SECTION G-G
SCALE: 4/1



SECTION K-K
SCALE: 4/1



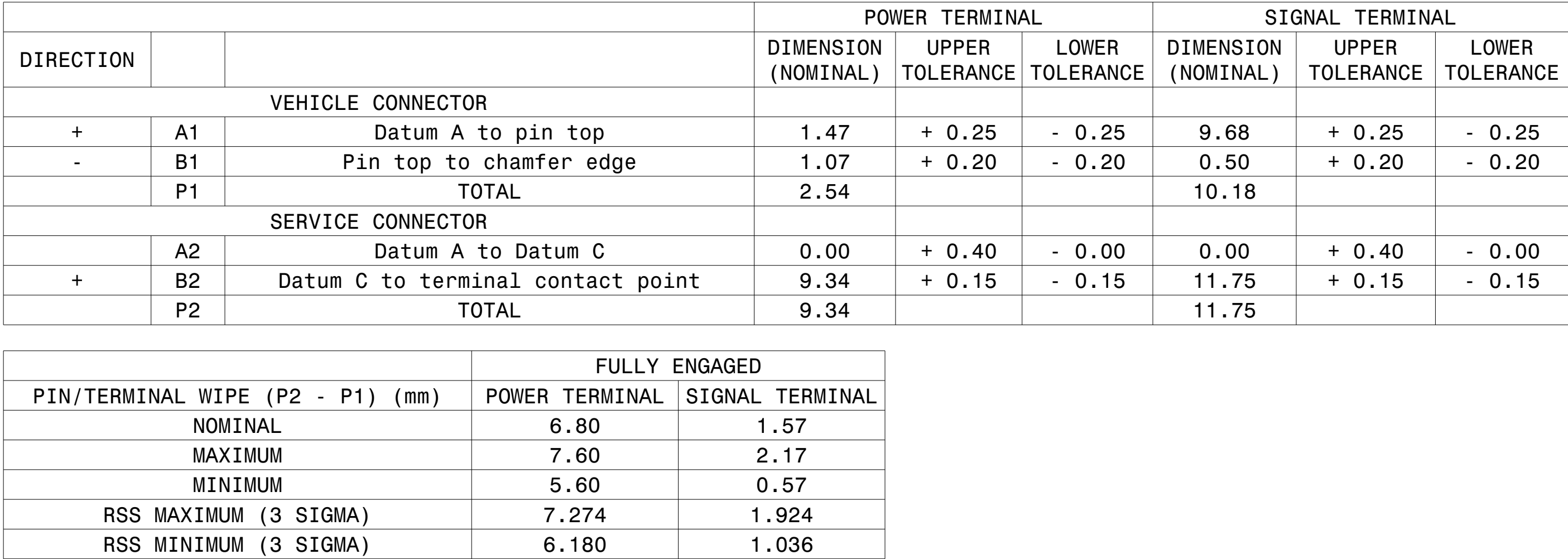
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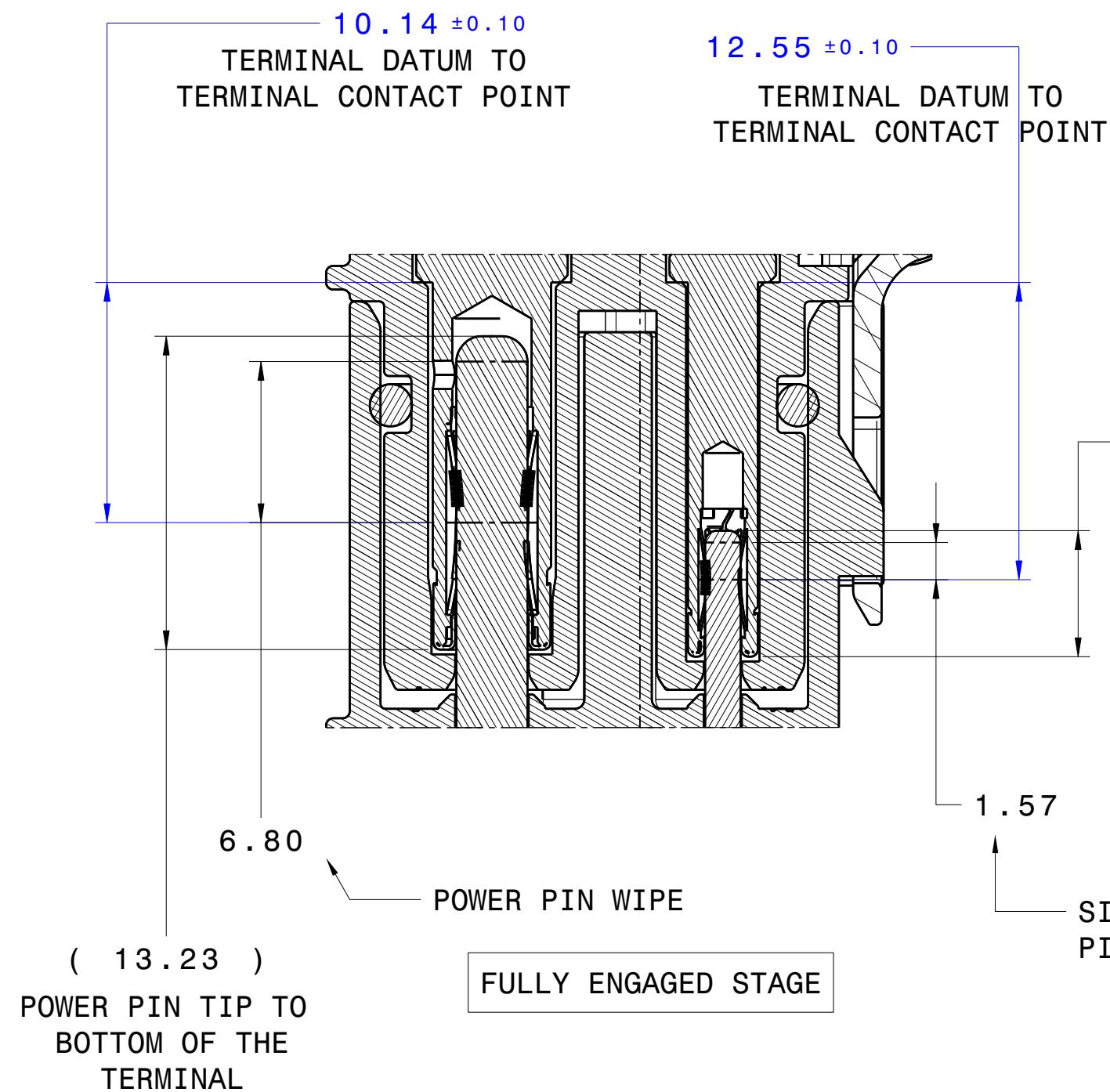
ITEM NAME
INTERFACE, MV EXTERNAL SUPPORT CONNECTOR

ITEM NUMBER	REVISION	SHEET
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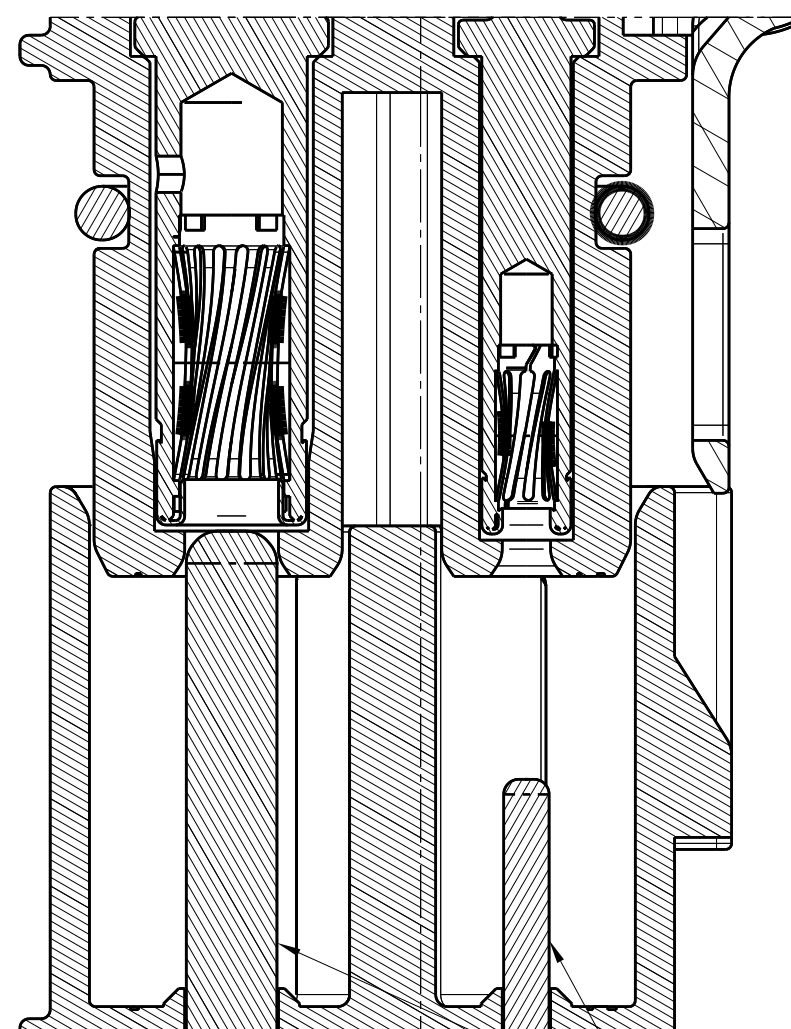


The image shows the Tesla logo at the top, consisting of the word "TESLA" in a stylized, black, sans-serif font. Below the logo is a block of text in a smaller, black, sans-serif font. The text reads: "THIS DOCUMENT IS PROPERTY OF TESLA INC. THE INFORMATION CONTAINED HEREIN IS TO BE KEPT CONFIDENTIAL, PROPRIETARY, AND A TRADE SECRET OF TESLA INC. THIS INFORMATION MAY NOT BE USED, REPRODUCED, OR DISCLOSED AS THE DIRECT OR INDIRECT BASIS FOR THE DEVELOPMENT, MANUFACTURE, OR SALE OF PROCESSES OR PRODUCTS WITHOUT THE EXPRESSED WRITTEN CONSENT OF TESLA INC." Below this text is a table with a black border. The table has two rows and two columns. The first row has headers "ITEM NAME" and "REVISION SHEET". The second row has values "INTERFACE, MV EXTERNAL SUPPORT CONNECTOR", "A.2", and "7 of 15".

ENGAGEMENT/DISENGAGEMENT SEQUENCE



FULLY ENGAGED STAGE



13.43 MM VERTICAL TRAVEL
FULLY DISENGAGED STAGE

POWER PIN AND SIGNAL
PIN OUT SIDE THE TERMINAL

10.14 ±0.10
TERMINAL DATUM TO
TERMINAL CONTACT POINT

(7.71)
POWER PIN TIP TO
BOTTOM OF THE
TERMINAL

1.28

POWER PIN WIPE

5.52 MM VERTICAL TRAVEL
PARTIALLY ENGAGED STAGE

VARIATION OF POWER PIN TIP FROM DATUM A: +/- 0.25 MM
VARIATION OF SIGNAL PIN TIP FROM POWER PIN TIP: +/- 0.2 MM
VARIATION OF CONTACT POINT ON POWER TERMINAL FROM
DATUM C: +/- 0.15 MM
VARIATION OF CONTACT POINT ON POWER TERMINAL FROM
CONTACT POINT ON SIGNAL TERMINAL: +/- 0.2 MM

TOTAL VARIATION ON POWER PIN WIPE WHEN SIGNAL PIN IS
OUTSIDE THE SIGNAL TERMINAL: = 0.25 + 0.2 + 0.15 + 0.2 = +/- 0.8 MM

WORST CASE POWER PIN WIPE (WHEN SIGNAL PIN IS OUTSIDE
THE SIGNAL TERMINAL): 1.28 - 0.8 = 0.48 MM
WORST CASE (RSS): 1.28 - 0.36 = 0.92 MM

12.55 ±0.10
TERMINAL DATUM TO
TERMINAL CONTACT POINT

SIGNAL PIN OUTSIDE
SIGNAL TERMINAL



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	10	9	8	7	6	5	4	3	Project:	VARIOUS	
H	GENERAL REQUIREMENTS 1. THE CONNECTORS SHALL HAVE A 2P+2S (2 POWER + 2 SIGNAL) CONFIGURATION. 2. THE VEHICLE CONNECTOR SHALL BE CONFIGURED TO BE A WIRE-TO-WIRE CONNECTOR. 3. THE VEHICLE CONNECTOR SHALL BE DESIGNED TO BE ASSEMBLED AT A TIER-1 HARNESS SUPPLIER. 4. THE VEHICLE CONNECTOR SHALL HAVE A CAP ATTACHED TO THE CONNECTOR BODY WHEN THE CONNECTOR IS NOT IN USE. 5. THE VEHICLE CONNECTOR SHALL HAVE A 11 MM CLIP SLOT AS DEFINED BY USCAR/EWCAP-005-11. 6. THE CONNECTORS SHALL ACCEPT VOLTAGE INPUT (UP-TO 60V) IN ORDER TO SUPPORT POWER SOURCES THAT ARE COMMON AND AVAILABLE TO ROADSIDE SERVICE AND CUSTOMERS. 7. POSITIVE AND NEGATIVE LOCATIONS ON BOTH THE CONNECTORS MUST BE CLEARLY MARKED. 8. THE SIGNAL PINS SHALL FOLLOW FIRST BREAK/LAST MATE. DURING PARTIAL ENGAGEMENT, THE POWER TERMINALS SHOULD HAVE AT LEAST 0.8 MM PIN WIPE (MINIMUM RSS) WHEN THE SIGNAL PINS ARE OUTSIDE THE SIGNAL TERMINAL. 9. THE CONNECTOR SHALL NOT BE ABLE TO BE PLUGGED IN A CONFIGURATION DIFFERENT THAN INTENDED (PROTECTED AGAINST REVERSE POLARITY). 10. THE CONNECTORS SHALL BE CONNECTED BY A MECHANICAL LATCH WHEN MATED TO PREVENT ACCIDENTAL UNPLUGGING WHILE IN USE. 11. MAXIMUM 0.4 MM OF AXIAL PLAY ALLOWED ON SERVICE CONNECTOR WHEN IT IS MATED TO THE VEHICLE CONNECTOR WITH A MECHANICAL LATCH. 12. PACKAGING SIZE OF INDIVIDUAL CONNECTORS MUST NOT EXCEED THE DIMENSIONS SHOWN ON SHEET 1. PACKAGING SIZE OF MATED PAIR MUST NOT EXCEED THE DIMENSIONS SHOWN ON SHEET 3. 13. POWER TERMINAL COMPATIBLE WIRE: 11-12 AWG SIGNAL TERMINAL COMPATIBLE WIRE: 19-20 AWG 14. UNDER MAXIMUM POWER LOADING AT 55 C AMBIENT, THE CONNECTOR HOUSING SURFACE TEMPERATURE SHALL NOT EXCEED 70 C.										H
G											G
F											F
E	CYCLE RATING VEHICLE CONNECTOR: MINIMUM 1,000 CYCLES SERVICE CONNECTOR: MINIMUM 10,000 CYCLES REQUIREMENT IS NOT FOR A MATED PAIR BUT FOR INDIVIDUAL CONNECTORS. SEALING PERFORMANCE 1. THE MATED CONNECTORS SHALL HAVE IP67 RATING. 2. UNMATED VEHICLE CONNECTOR (WITH PROTECTIVE CAP INSTALLED) SHALL HAVE IP69K RATING. 3. UNMATED VEHICLE CONNECTOR (WITH PROTECTIVE CAP REMOVED) SHALL HAVE IP67 RATING. 4. UNMATED SERVICE CONNECTOR SHALL HAVE IP67 RATING.										E
D											D
C	ERGONOMICS 1. SIZE REQUIREMENTS FOR FINGER GRIP FEATURES (SEE APPENDIX SHEETS FOR DETAILS) A - FINGER GRIP FEATURE ON SERVICE CONNECTOR LATCH: MINIMUM 8 MM X 9 MM B - FINGER GRIP FEATURES ON CONNECTOR BODY (BOTTOM): UP TO SUPPLIER DISCRETION C - FINGER GRIP FEATURES ON CONNECTOR BODY (SIDES): UP TO SUPPLIER DISCRETION 2. DEPRESS FORCE ON LATCH SHALL NOT EXCEED 15N. 3. THE VEHICLE CONNECTOR SHALL HAVE A 11 MM CONNECTOR RETAINER SLOT AS DEFINED BY USCAR/EWCAP-005-11. 4. TERMINAL AND SEAL DESIGN SHALL BE DESIGNED FOR A MAXIMUM TOTAL ASSEMBLY FORCE OF </= 75N. 5. THE SERVICE SIDE CONNECTOR WIRING HARNESS SHALL BE ORIENTED 180° TO THE DIRECTION OF MATING.										C
B											B
A	COLOR THE CONNECTORS SHALL BE AVAILABLE IN LIGHT BLUE (RAL 5012 OR EQUIVALENT) IN COLOR. SERVICING PRY POINTS FOR REMOVAL OF ANY INTERNAL PARTS OF THE VEHICLE SIDE CONNECTOR, SUCH AS THE INNER HOUSING OF THE INLINE SIDE, MUST BE VISUALLY OBVIOUS. TERMINAL EXTRACTION ON VEHICLE SIDE CONNECTOR MUST NOT REQUIRE UNCOMMON TOOLS OR EQUIPMENT.										A
	10	9	8	7	6	5	4	3	2	1	SHEET SIZE C

109876543

Project:VARIOUS

H

CONNECTOR ELECTRICAL REQUIREMENTS

VOLTAGE RATING

NOMINAL VOLTAGE: 48V AC OR DC.
MAXIMUM VOLTAGE: 60V AC OR DC.
MAXIMUM TRANSIENT VOLTAGE: 330V AC OR DC.

CREEPAGE AND CLEARANCE

THE POWER TERMINALS FOR BOTH VEHICLE AND SERVIE CONNECTOR SHALL PASS IEC 60664-1
CREEPAGE AND CLEARANCE REQUIREMENT FOR NOMINAL AND MAX VOLTAGES IN ADDITION TO MEETING:
BASIC INSULATION RATING
ALTITUDE = 5000 M (CORRECTION FACTOR = 1.48)
POLLUTION DEGREE : 3 EQUIPMENT

	MATERIAL GROUP 1 (RECOMMENDED)	MATERIAL GROUP 2	MATERIAL GROUP 3
MIN. CREEPAGE (MM)	1.6	1.8	2.0
MIN. CLEARANCE (MM)	1.2	1.2	1.2

MATERIALS

1. ALL PLASTIC IN CONTACT WITH CONDUCTORS EXPOSED TO VOLTAGE SHOULD BE RATED TO
HB FLAMMABILITY PER UL 94.
2. ELECTRICAL RTI MUST MEET 125 C REQUIREMENT.
3. RECOMMENDED TO USE CTI (COMPARATIVE TRACKING INDEX EQUAL TO OR GREATER THAN 600 ,

CONTACT RESISTANCE

TESTING SHOULD BE DONE ACCORDING TO USCAR 2-8 SECTION 5.3.1 DRY CIRCUIT RESISTANCE PROCEDURE. THE CONNECTORS SHALL MEET
ACCEPTANCE CRITERIA ACCORDING TO USCAR 2-8 SECTION 5.3.2.4.

CONTINUOUS CURRENT

THE CONNECTOR PAIR MUST BE ABLE TO SUPPORT 30 A CONTINUOUS AT 55 C AMBIENT TEMPERATURE.

SIGNAL INTEGRITY

THE SIGNAL PINS SHALL SUPPORT CAN COMMUNICATION (500 KBPS) UNDER OPERATING ENVIRONMENT DESCRIBED ABOVE.

G

F

E

D

C

B

A

H

G

F

E

D

C

B

A

SHEET SIZE

C

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1

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Project:VARIOUS

H

GENERAL RELIABILITY REQUIREMENTS

04

1. ALL CONNECTORS MUST MEET APPLICABLE PERFORMANCE REQUIREMENTS OF USCAR-2.
TEMPERATURE CLASS: T2
SEALING CLASS: S3 - (2144149-XX)
S2 - (2144150-XX)
VIBRATION CLASS: V2

G

ADDITIONAL TESLA RELIABILITY REQUIREMENTS

1. ALL CONNECTORS MUST PASS THE TESLA RS-3690003 VERSION 2 RELIABILITY TEST SPECIFICATIONS.

2. ALL VEHICLE AND SERVICE CONNECTORS MUST ADDITIONALLY PASS TESLA RS-3690003 VERSION 2 SECTION 5.3 (HTOE) WITH THE FOLLOWING DEVIATIONS:
MATING CONDITION: UNMATED, CAP INSTALLED
OPERATING MODE: NO CURRENT
TEST DURATION: 4500 HOURS (SEE TESLA RS-3690003 TABLE 6).

3. ALL VEHICLE AND SERVICE CONNECTORS MUST ADDITIONALLY PASS TESLA RS-3690003 VERSION 2 SECTION 5.4 (HTHE) WITH THE FOLLOWING DEVIATIONS:
MATING CONDITION: UNMATED, CAP INSTALLED
OPERATING MODE: NO CURRENT
TEST DURATION: 925 HOURS (SEE TESLA RS-3690003 TABLE 8).

4. ALL VEHICLE CONNECTORS MUST ADDITIONALLY PASS TESLA RS-3690003 VERSION 2 SECTION 5.7 (VIBRATION AND THERMAL CYCLING) WITH THE FOLLOWING DEVIATIONS:
MATING CONDITION: UNMATED, CAP INSTALLED
OPERATING MODE: NO CURRENT
CONNECTOR INSTALLATION: REPRESENTATIVE OF INSTALL POSITION ON VEHICLE
CONNECTOR MOUNTING METHOD: 11 MM CLIP
NOTE: VEHICLE CLASS M3, VIBRATION CLASS V2.
TEST PASS CRITERIA:

1. VEHICLE CONNECTOR CAP REMAINS INTACT

2. VEHICLE CONNECTOR (WITH CAP INSTALLED) RETAINS IP69K PROTECTION.

5. ALL VEHICLE CONNECTORS MUST ADDITIONALLY PASS TESLA RS-3690003 VERSION 2 SECTION 5.8 AND 5.9 (MECHANICAL SHOCK – POTHOLE AND COLLISION) WITH THE FOLLOWING DEVIATIONS:
MATING CONDITION: UNMATED, CAP INSTALLED
OPERATING MODE: NO CURRENT
CONNECTOR INSTALLATION: REPRESENTATIVE OF INSTALL POSITION ON VEHICLE
CONNECTOR MOUNTING METHOD: 11 MM CLIP
ADDITIONAL TEST DISPOSITION CRITERIA: VEHICLE SIDE CONNECTOR CAP IN IT’S NOMINAL POSITION (ATTACHED TO THE CONNECTOR) AFTER THE TEST.
NOTE: VEHICLE CLASS M4, VIBRATION CLASS V2.
TEST PASS CRITERIA:

1. VEHICLE CONNECTOR CAP REMAINS INTACT

2. VEHICLE CONNECTOR (WITH CAP INSTALLED) RETAINS IP69K PROTECTION.

6. ALL VEHICLE AND SERVICE SIDE CONNECTORS MUST ADDITIONALLY PASS TESLA RS-3690003 VERSION 2 SECTION 5.13 (CHEMICAL EXPOSURE) WITH THE FOLLOWING DEVIATIONS:
DUT: APPLIES TO UNMATED VEHICLE SIDE CONNECTOR (CAP INSTALLED), UNMATED SERVICE SIDE CONNECTOR, AND MATED PAIR OF VEHICLE SIDE AND SERVICE SIDE CONNECTORS.

F

E

D

C

B

A

SHEET SIZE

C

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DOCUMENTATION

SUPPLIERS MUST PROVIDE 2 CLEAR, SEPARATE TEST REPORTS SHOWING:

1. USCAR VALIDATION EITHER IN ITS ENTIRETY OR AS REQUIRED BY TESLA RS-3690003. THIS DOCUMENT IS UNCONTROLLED BY TESLA.
2. TESLA RS-3690003 VALIDATION UNIQUE TO TESLA FOR THE TESTS DESCRIBED IN SECTION 5. THIS DOCUMENT SHALL NOT BE SHARED WITH ANY THIRD PARTY UNLESS AUTHORIZED BY TESLA.

SEAL ANALYSIS DOCUMENTATION

PRIOR TO PRODUCTION OF ANY NEW SEALS (RADIAL, SINGLE WIRE, ETC.), SUPPLIERS MUST PROVIDE DETAILED SEAL PERFORMANCE ANALYSIS DOCUMENTATION APPROVED BY TESLA. THIS SHOULD INCLUDE, BUT IS NOT LIMITED TO:

1. CONTACT PRESSURE, COMPRESSION AMOUNT (MM), SEAL STRETCH, INSTALL AND REMOVAL FORCES, STRESS, STRAIN, AND VOLUME FILL AT THE GEOMETRY'S LEAST MATERIAL CONDITION, NOMINAL, AND MAXIMUM MATERIAL CONDITIONS.
2. DYNAMIC FEA OF THE SEAL MATING WITH THE INTERFACE INSPECTING FOR SEAL ROLLING AT THE LEAST MATERIAL CONDITION, NOMINAL, AND MAXIMUM MATERIAL CONDITIONS.

PRODUCTION TESTING REQUIREMENTS

ELECTRICAL ISOLATION

ISOLATION RESISTANCE:

THE ASSEMBLY SHALL MEET A MINIMUM ISOLATION RESISTANCE OF ≥ 100 M-OHM AT MIN. 330V DC BETWEEN ADJACENT CONDUCTORS.

HIPOT TESTING:

THE ASSEMBLY SHALL WITHSTAND THE TEST VOLTAGE BETWEEN ADJACENT CONDUCTORS RAISED UNIFORMLY FROM 0V TO 330 V DC AT 23 ±5 °C.

TO SHORTEN TEST TIMES, LEAKAGE CURRENT LIMITS DO NOT NEED TO BE ENFORCED DURING CHARGING/RAMP UP OF THE TEST ARTICLE. THE TESTER MUST BE ABLE TO DETECT GROSS BREAKDOWN (CURRENTS GREATER THAN 30MA) DURING ALL PHASES OF THE TEST, INCLUDING DURING CHARGING/RAMP UP OTHERWISE THIS WILL NEED TO BE COMPENSATED FOR WITH SLOWER CHARGING/RAMP UP.

REQUIREMENTS:

THE ASSEMBLY MUST MEET THE ACCEPTANCE REQUIREMENT OF SAE J1742, SECTION 5.5.2.4:

"THERE SHALL BE NO DIELECTRIC BREAK-DOWN OR FLASH-OVER BETWEEN INDIVIDUAL CAVITIES OR BETWEEN CAVITIES AND THE OUTSIDE OF THE CONNECTOR AT ANY TIME DURING THE TEST."

ELECTRICAL RESISTANCE

END-TO-END RESISTANCE LESS THAN OR EQUAL TO SUM OF APPLICABLE CONDUCTOR RESISTANCES SPECIFIED BY ISO 6722 AND TERMINAL SPECIFICATIONS.

PRESSURE TEST (MATED CONNECTORS AND VEHICLE SIDE CONNECTORS WITH CAP REMOVED)

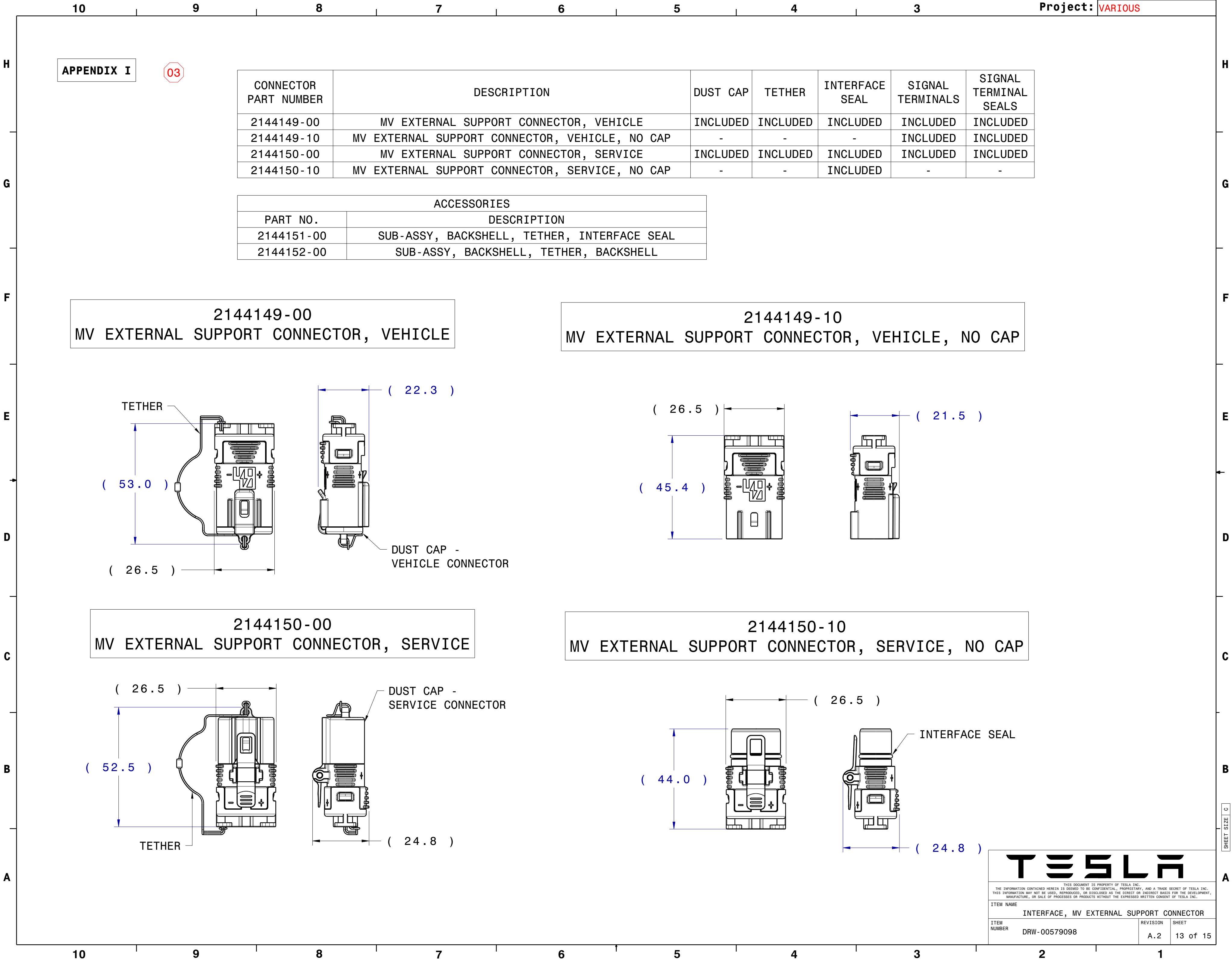
- TEST PRESSURE: +1.5 PSI INTERNALLY PRESSURIZED
- ACCEPTANCE CRITERIA: <0.03 PSI PRESSURE LOSS
- TEST TIME: 3 SECONDS
- TESTER VOLUME (INCLUDING TUBING): <1.8E-5 M3



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ITEM NAME	INTERFACE, MV EXTERNAL SUPPORT CONNECTOR
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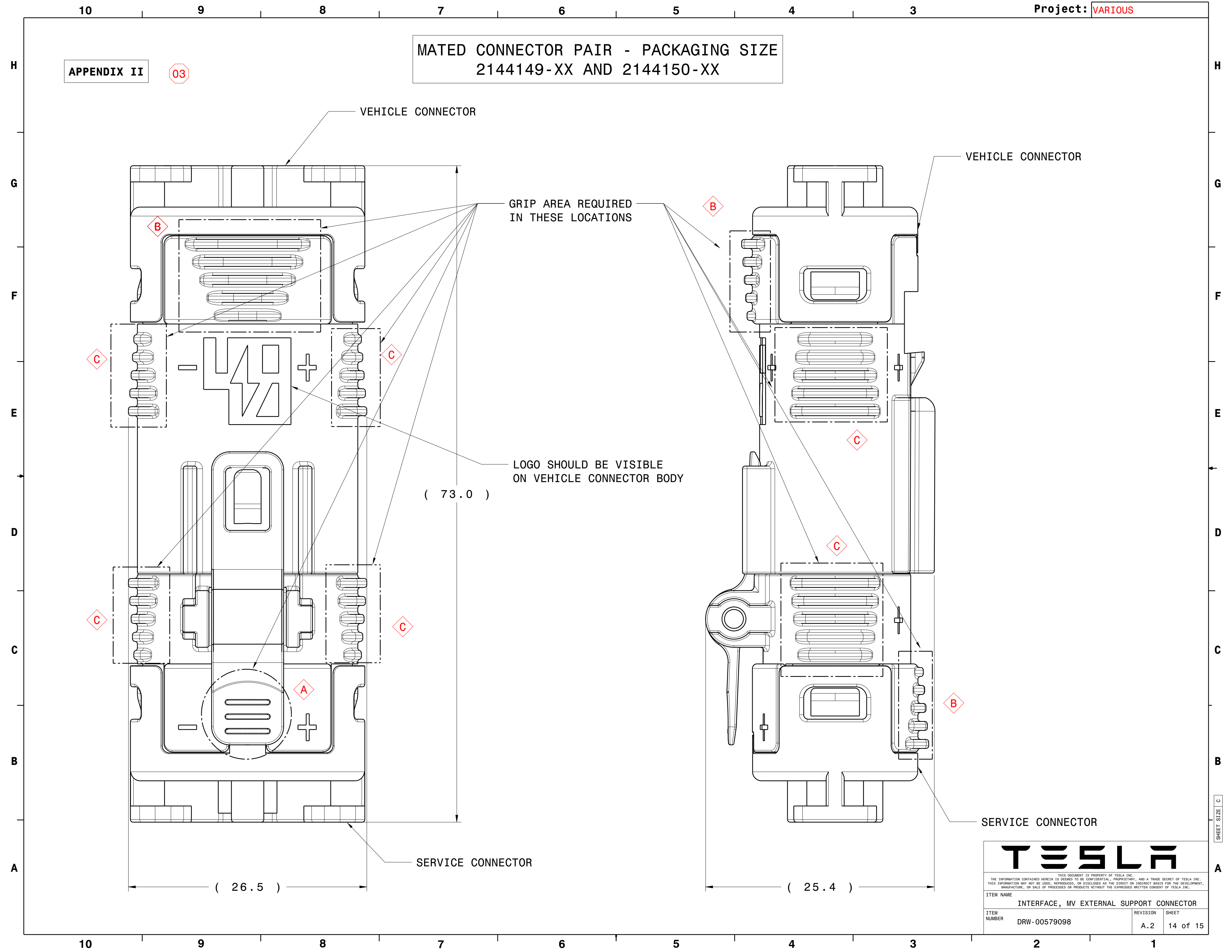
ITEM NUMBER	DRW-00579098	REVISION A 2	SHEET 12 of 15
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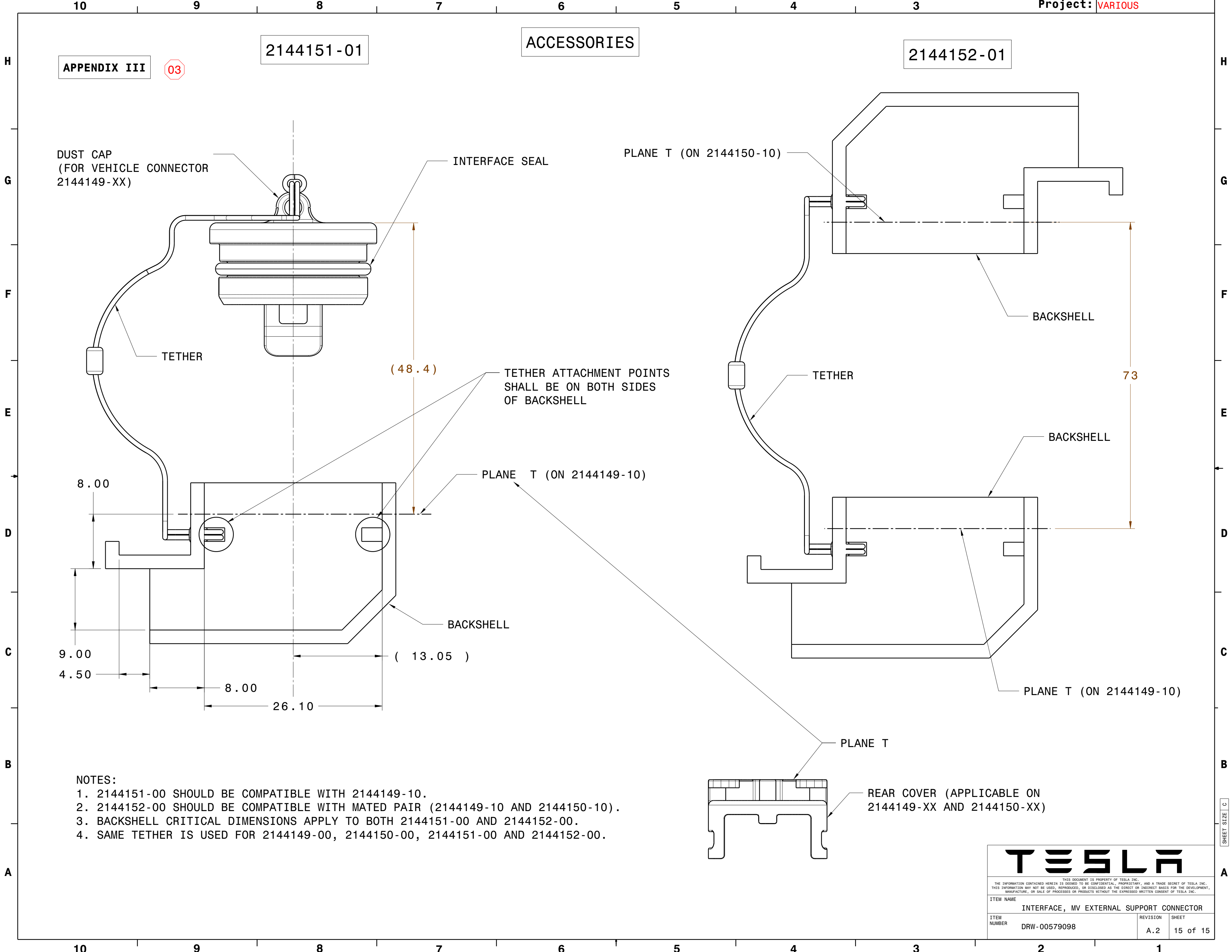


APPENDIX II

03

MATED CONNECTOR PAIR - PACKAGING SIZE
2144149-XX AND 2144150-XX





APPENDIX III

03

2144151-01

ACCESSORIES

2144152-01

Project: VARIOUS

DUST CAP
(FOR VEHICLE CONNECTOR
2144149-XX)

INTERFACE SEAL

PLANE T (ON 2144150-10)

TETHER

(48 . 4)

TETHER ATTACHMENT POINTS
SHALL BE ON BOTH SIDES
OF BACKSHELL

PLANE T (ON 2144149-10)

BACKSHELL

(13 . 05)

TETHER

BACKSHELL

73

BACKSHELL

PLANE T (ON 2144149-10)

PLANE T

REAR COVER (APPLICABLE ON
2144149-XX AND 2144150-XX)

- NOTES:
1. 2144151-00 SHOULD BE COMPATIBLE WITH 2144149-10.
 2. 2144152-00 SHOULD BE COMPATIBLE WITH MATED PAIR (2144149-10 AND 2144150-10).
 3. BACKSHELL CRITICAL DIMENSIONS APPLY TO BOTH 2144151-00 AND 2144152-00.
 4. SAME TETHER IS USED FOR 2144149-00, 2144150-00, 2144151-00 AND 2144152-00.

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ITEM NAME
INTERFACE, MV EXTERNAL SUPPORT CONNECTOR

ITEM NUMBER
DRW-00579098

REVISION
A.2

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